MERAKI - BIOS : RE-NEET 2026
TARGET 360/360 IN BIOLOGY
THREE [3] ALL INDIA FREE TESTS IN BIOLOGY
27.05.2026; 07.06.2026, 17.06.2026

PRACTICE TEST: REVISE AND RECALL: PLANT GROWTH AND DEVELOPMENT

1. Consider the following statements regarding plant growth:

I. Growth in plants involves a permanent and irreversible increase in size.
II. All cells produced by meristems continue to divide throughout the life of the plant.
III. Plant growth is called open growth because new cells are continuously added by meristems.

Which of the above statements are correct?

1. I only
2. I and II only
3. I and III only
4. I, II and III

[Page 167 - Recall - Cellular totipotency in plants and concept of micropropagation]

2. During differentiation, to form a tracheary element, the cells in a plant:

A: would lose their protoplasm.

B: develop a very strong, lignocellulosic secondary cell wall.

1. Only A is correct
2. Only B is correct
3. Both A and B are correct
4. Both A and B are incorrect

[Page 172 - Recall - Differentiation, Dedifferentiation and Redifferentiation]

3. Which of the following best explains the phenomenon of heterophylly observed in buttercup?

1. The plant produces different leaf shapes at juvenile and mature stages due to internal genetic programming
2. The plant develops distinct leaf forms in water versus air, demonstrating environmentally induced plasticity
3. The plant undergoes mutation in response to changing seasons, altering its leaf morphology permanently
4. The plant mimics the leaf shape of neighboring species as an adaptive survival mechanism

[Page 157 - Recall - is plasticity more important for plants or for animals?]

4. Which of the following PGR are adenine derivatives?

1. Auxins
2. Cytokinins
3. Gibberellins
4. Abscisic acid

[Page 174; Recall the chemical nature of all PGR]

5. Identify the correct statement/s regarding auxins?

A: They are generally produced by the growing apices of the stems and roots, from where they migrate to the regions of their action.

B. Auxins like NAA (naphthalene acetic acid) and 2, 4-D (2, 4- dichlorophenoxyacetic) have been isolated from plants.

C. IAA and indole butyric acid (IBA) are synthetic auxins.

1. Only B and C
2. A, B and C
3. Only A
4. None

[Page 175; Recall - Uses of auxins in Horticulture]

6. Consider the given two statements:

Assertion [A]: Spraying sugarcane crop with gibberellins increases the yield by as much as 20 tonnes per acre.

Reason [R]: Spraying sugarcane crop with gibberellins increases the length of the stem.

1. Both A and R are true and R explains A
2. Both A and R are true but R does not explain A
3. A is true but R is false
4. A is false but R is true

[Page 176; Recall - all functions of GAs - very commonly asked in NEET]

7. Which of the following is/are function/s of cytokinins in plants?

I. Cytokinins help to produce new leaves, chloroplasts in leaves, lateral shoot growth and adventitious shoot formation.

II. Cytokinins help overcome the apical dominance.

III. Cytokinins promote nutrient mobilisation which helps in the delay of leaf senescence.

1. Only I and II
2. Only I and III
3. Only II and III
4. I, II and III

8. In most situations, which PGR acts as an antagonist to GAs?

1. Auxins
2. ABA
3. Ethylene
4. Cytokinins

[Page 178; Recall - What is the role of ABA in seed dormancy and germination?]

9. How many of the given functions are performed by ethylene in plants?

A: Horizontal growth of seedlings, swelling of the axis and apical hook formation in dicot seedlings

B: Promotes senescence and abscission of plant organs especially of leaves and flowers.

C: Enhances the respiration rate during ripening of the fruits.

1. 0
2. 1
3. 2
4. 3

[Page 177 - Recall - Read all about ethylene now - very important for NEET]

10. Consider the given two statements:

I: The most widely used compound as source of ethylene is ethephon.

II: Ethephon in an aqueous solution is readily absorbed and transported within the plant and releases ethylene slowly.

1. Both I and II are correct.
2. Only I is correct.
3. Only II is correct
4. Both I and II are incorrect.